

A Privacy-by-Design Framework for Chatbot Conversation Anonymisation with Hybrid NLP and Bootstrap-Based Risk Assessment

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Abstract

This paper presents a privacy-by-design framework for anonymising multilingual chatbot conversations collected at the Italian National Institute of Statistics (Istat), within contact centre and user support services. These interactions generate large volumes of unstructured text containing Personally Identifiable Information (PII), requiring scalable solutions that ensure privacy while preserving analytical usability. We propose a hybrid anonymisation pipeline combining rule-based detection of structured identifiers with a fine-tuned multilingual Named Entity Recognition (NER) model based on a BERT architecture. This approach enables robust identification of both structured and context-dependent PII in heterogeneous conversational data. The proposed approach is evaluated on a manually curated gold standard of real conversations. A bootstrap-based evaluation strategy is used to assess performance under uncertainty. Results show high sensitivity and low information loss, indicating an effective balance between privacy protection and data preservation. We further introduce a conversation-level indicator of residual disclosure risk, which aggregates token-level errors into an interpretable measure for operational monitoring. This provides a practical tool for supporting privacy management in production environments. Overall, the study shows that combining NLP-based extraction with statistical risk assessment enables scalable and interpretable anonymisation of conversational data in official statistics, supporting both compliance and downstream text analysis.

Keywords: Named Entity Recognition, Anonymisation, Disclosure Risk.

1. Introduction

Digital contact channels are widely used by National Statistical Institutes (NSIs) to provide assistance and disseminate information. Within public administration, chatbots have become a standard tool for service delivery and rank among the most prominent Artificial Intelligence (AI) applications in the public sector (European Commission, 2020; van Noordt & Misuraca, 2022). At the Italian National Institute of Statistics (Istat), these conversational agents support contact centre operations and enable users to query statistical services through natural language, generating large volumes of textual data with significant analytical potential.

However, chatbot conversations are unstructured, multilingual, and frequently contain Personally Identifiable Information (PII) provided by users. This raises a key challenge: how to exploit their analytical utility while ensuring strict privacy protection and compliant data storage. In the domain of official statistics, this requires extending Statistical Disclosure Control (SDC) to unstructured textual data, in line with privacy-by-design principles and the General Data Protection Regulation (GDPR) (Istat, 2025).

The literature on text anonymisation remains divided between NLP approaches focused on entity detection and privacy-preserving data publishing methods centred on formal risk

modelling. Their integration into a unified framework remains an open issue, particularly for such conversational data.

This study proposes a scalable and reproducible system for anonymising chatbot conversations at Istat. The approach combines NLP-based extraction techniques with inferential risk assessment methods. Specifically, a hybrid anonymisation pipeline integrates deterministic rules for structured identifiers with a fine-tuned multilingual NER model based on a BERT architecture. In addition, a statistically grounded evaluation strategy employs a bootstrap-based inferential approach to obtain robust estimates—along with confidence intervals—for sensitivity, residual disclosure risk, and information loss. The proposed approach also introduces a conversation-level indicator of residual disclosure risk, providing a practical tool for assessing the privacy–utility trade-off at the level of entire dialogue sessions.

Pseudonymisation and masking are considered implementation strategies within the anonymisation process, while the focus is on PII detection and the statistical evaluation of residual risk. Consistent with SDC terminology, de-identification is treated as the technical extraction phase within a comprehensive anonymisation framework.

The remainder of the paper is structured as follows. Section 2 reviews the theoretical background on privacy-preserving NLP. Section 3 details the data sources and the construction of the synthetic training corpus. Section 4 formalises the proposed approach and the inferential risk assessment methodology. Section 5 presents the empirical results and discusses their implications for statistical text analysis. Section 6 concludes.

2. Privacy-preserving NLP: background and related work

2.1 NLP-based approaches to text anonymisation

The anonymisation of unstructured data has been extensively examined in the privacy-preserving data mining literature. Unlike structured microdata, free text may contain latent identifiers and contextual clues that are difficult to detect automatically (Dernoncourt et al., 2017; Lison et al., 2021).

Early approaches relied on rule-based methods, in particular regular expressions, which achieve high precision in detecting structured identifiers (e.g. tax codes) but are less effective for context-dependent PII. A foundational contribution is provided by Medlock (2006), who introduced a probabilistic approach to anonymisation in e-mail and highlighted the distinction between direct identifiers (e.g., personal names) and quasi-identifiers (e.g., organisations or contextual references). However, this line of work mainly focuses on detection performance rather than disclosure risk.

Subsequent studies confirm the limitations of purely deterministic methods in heterogeneous textual settings (Jaikumar and Mohana, 2023). Recent work has therefore shifted towards deep learning models, particularly Transformer-based architectures such as BERT, which leverage contextual representations to improve entity recognition (Al-Sabbagh et al., 2025).

Hybrid approaches combining rule-based and neural methods are now commonly adopted, providing more robust performance across diverse datasets (Mishra et al., 2025). In this context, Named Entity Recognition (NER) is typically formulated as a sequence labelling task (Chiu and Nichols, 2016; Lample et al., 2016).

A large part of the literature focuses on the clinical domain, where the objective is the detection of Protected Health Information (PHI) in medical texts (Meystre et al., 2010; Aberdeen et al.,

2010). While these studies provide benchmarks and evaluation frameworks, their domain specificity limits transferability to multilingual and open-ended conversational data.

2.2 Risk-based anonymisation and privacy models

In parallel, the literature on privacy-preserving data publishing (PPDP) approaches anonymisation as a risk-driven transformation process under formal privacy constraints. The objective is not only to remove identifiers but also to quantify and control disclosure risk.

As discussed by Sánchez and Batet (2016), anonymisation can be formalised as a transformation that satisfies predefined privacy guarantees while preserving data utility. Similarly, El Emam et al. (2015) show that residual disclosure risk may persist even after removing explicit identifiers, due to quasi-identifiers and rare attribute combinations.

These approaches are mainly developed for structured data, where risk can be modelled using well-defined probabilistic frameworks. Their extension to unstructured text remains limited, due to high dimensionality, linguistic variability, and contextual ambiguity.

As a result, two largely separate paradigms can be identified: (i) NLP-based methods, focused on entity detection and extraction performance; (ii) PPDP approaches, focused on formal risk modelling and privacy guarantees. Their integration remains an open issue.

2.3 Research gap and contribution

Existing approaches do not simultaneously address accurate PII detection, preservation of analytical utility, and quantitative assessment of residual disclosure risk in conversational settings. This limitation is particularly relevant for such contexts, where linguistic variability and contextual ambiguity increase both detection complexity and re-identification risk.

In practice, most contributions focus either on entity recognition performance or on disclosure risk modelling under simplified assumptions, with limited integration between these perspectives.

This study addresses this gap by proposing a unified framework for the chatbot data anonymisation at Istat. The approach combines a hybrid detection pipeline, based on rule-based detection and transformer-based NER, with a bootstrap-based inferential evaluation strategy. A conversation-level indicator of residual disclosure risk is also introduced, enabling assessment at the level of entire dialogues.

3. Data Sources and Dataset Construction

The empirical analysis is based on chatbot conversations collected at Istat between November 10, 2025 and March 31, 2026. The data originate from two main digital contact channels: the Contact Centre, dedicated to citizen support, and the Business Portal (Statistica & Imprese), used by enterprises for data collection management.

The dataset consists of 69,627 conversations, of which 27,801 (39.92%) include user-generated messages extracted from daily JSON-formatted conversational logs. Each conversation is structured as a sequence of messages enriched with metadata (e.g. sender type and textual content), preserving the full interaction structure. The distribution across channels is relatively

balanced, with 39,312 messages (56.46%) from the Business Portal¹ and 30,315 messages (43.54%) from the Contact Centre.

The corpus of user-generated messages contains 8,733 unique word types, reflecting the linguistic variability typical of digital public services interactions. The data are predominantly Italian (68.81%), while 11.41% of messages are classified as undefined due to mixed language usage, short texts or insufficient linguistic cues. Among the remaining languages, Portuguese accounts for 5.11%, followed by English (4.99%) and German (1.90%).

Although the anonymisation process primarily targets user-generated messages, chatbot responses may also contain PII, as systems can replicate or reformulate sensitive information. For this reason, PII detection is applied to the full conversational content. The corpus includes a wide range of PII categories, from structured identifiers (e.g. email addresses, protocol numbers) to numerical identifiers, including phone numbers, VAT numbers, as well as context-dependent elements such as user IDs and license plates, and more generic patterns such as dates.

In addition to the raw data, a synthetically generated dataset is used for training the NER component. The synthetic corpus is designed to replicate typical linguistic structures observed in real interactions while ensuring full compliance with privacy constraints. The generation process relies on approximately 50 sentences, including both PII-containing and negative examples without sensitive information. These templates are populated with realistic entities drawn from authoritative sources. In particular, personal names and surnames are sampled from large-scale Italian onomastic repositories (approximately 9,000 first names and 20,000 surnames), while address components are derived from the National Register of Street Names and Civic Numbers, ensuring realistic geographic variability based on officially updated spatial information.

The resulting dataset is structured in CoNLL format (Tjong Kim Sang and De Meulder, 2003) using the BIO tagging scheme. This approach ensures annotation consistency and enables scalable dataset construction without manual labelling. The dataset exhibits a controlled distribution of entities: 28.07% of tokens are labelled as entities, with addresses slightly more frequent than personal names (55.84% vs 44.16%). At the sentence level, 95.6% of instances contain at least one entity, providing a dense training signal.

Despite these advantages, the use of synthetic data introduces limitations in terms of generalisation. In particular, the controlled generation process does not fully capture the noise, ambiguity, and linguistic variability of real chatbot interactions, such as spelling errors, code-switching, fragmented syntax, or implicit contextual references. As a result, models trained on synthetic data may exhibit reduced performance when applied to real-world data, especially for context-dependent or weakly structured PII.

To mitigate this risk, the synthetic corpus is designed to maximise variability within templates and entity combinations, while the evaluation is conducted on real conversational data annotated through a manually curated gold standard (Section 4). This separation between training (synthetic) and evaluation (real) allows us to assess the robustness of the model under realistic conditions. In addition, the hybrid architecture contributes to improving generalisation by capturing both structured patterns and contextual variability.

¹ The chatbot service was subsequently integrated into the Business Portal, with deployment starting on February 9, 2026.

Overall, while the synthetic dataset ensures privacy compliance and annotation control, the proposed framework explicitly accounts for its limitations by incorporating evaluation procedures that quantify performance degradation and residual disclosure risk in real conversational settings.

4. Methodology

The proposed anonymisation framework is based on a modular Python pipeline designed to ensure reproducibility and scalability. The system operates on JSON-formatted chatbot conversations, producing anonymised output while preserving the original conversational structure. The approach follows a hybrid design combining two complementary components: rule-based detection of structured identifiers through regular expressions and a fine-tuned transformer-based NER model for unstructured entities. PII is replaced with contextual placeholders (such as [NAME], [DATE], [ADDRESS], [NUMBER], [EMAIL], etc.), preserving textual readability and analytical usability.

4.1 Formalisation: PII Detection as a Binary Task

The identification of sensitive information is formalised as a binary classification problem at the token or entity-span level. Let x_i denote the i -th textual unit, and $y_i \in \{0,1\}$ its true label, where $y_i = 1$ indicates the presence of PII. The predicted label is $\hat{y}_i$. Performance is evaluated against a manually annotated gold standard using TP, FP, TN, and FN. We compute:

- Sensitivity (Recall): $\frac{TP}{(TP+FN)}$, representing the model's ability to capture all sensitive information.
- Precision: $\frac{TP}{(TP+FP)}$, indicating the reliability of the anonymisation.
- False Negative Rate (FNR): $\frac{FN}{(FN+TP)}$, which directly relates to the residual disclosure risk.
- Information Loss (False Positive Rate): $\frac{FP}{(FP+TN)}$, which measures the proportion of non-sensitive tokens erroneously masked, representing the impact on textual utility.

Given the strong class imbalance, evaluation focuses on precision–recall trade-offs rather than ROC-based metrics. This reflects an explicit trade-off between disclosure risk and information utility. Increasing sensitivity reduces false negatives and thus residual disclosure risk, but may increase over-masking and reduce textual utility. From an operational perspective, this corresponds to a conservative anonymisation setting where privacy protection is prioritised over maximal data preservation, in line with official statistics requirements.

To account for sampling variability, a non-parametric bootstrap procedure (1,000 replicates) is used to derive confidence intervals for performance metrics, treating them as random variables rather than fixed quantities.

4.2 Rule-Based Detection of Structured Identifiers

The first stage of the pipeline relies on deterministic pattern-matching rules implemented through regular expressions to identify and mask structured PII. These patterns are designed to capture well-defined identifiers commonly found in conversational interactions, including

email addresses, Italian fiscal codes, VAT numbers, IBANs, phone numbers, user identifiers and account codes, explicitly provided passwords, vehicle license plates, and dates expressed in both numerical and textual formats. Particular attention is given to ambiguous numeric patterns and vehicle license plates, which are not anonymised indiscriminately but only when occurring within a relevant semantic context. This contextual activation is implemented through the presence of lexical cues such as “user”, “code”, or “vehicle”, which trigger the application of masking rules. This mechanism reduces the risk of false positives arising from sequences of characters that may resemble identifiers but do not actually correspond to PII. This context-aware design reduces over-masking and improves the precision of the rule-based component, particularly in heterogeneous conversational settings where similar alphanumeric patterns may occur in non-sensitive contexts.

4.3 Named Entity Recognition with Fine-Tuned BERT

The second stage of the pipeline addresses unstructured personal information through a transformer-based NER model. This component is based on the Multilingual BERT (mBERT) case model, specifically fine-tuned as a token classification task on a domain-specific dataset. The training set was formatted according to the CoNLL standard and manually labelled to identify relevant entity categories, including person names and locations. The model is implemented using the Hugging Face Transformers library, which utilises a sub-word tokenisation strategy. The fine-tuning process was configured for five epochs; however, we implemented an early stopping strategy to monitor the validation loss and prevent overfitting. This ensured that the training procedure was terminated at the optimal convergence point, preserving the model's ability to generalise from the synthetic training patterns to the linguistic nuances of real-world interactions. During inference, the model predicts entity spans at the token level within each message. These predictions are then aligned and replaced with standardised placeholders, ensuring consistent anonymisation across the corpus.

4.4 Hybrid Anonymisation Strategy

The pipeline adopts a sequential logic to balance computational precision with linguistic flexibility. In the first stage, high-precision regular expressions are used to capture well-defined structured identifiers, thereby reducing the noise in the input space for the neural model. Subsequently, a transformer-based NER model is applied to detect residual unstructured entities that exhibit high contextual and linguistic variability. This hybrid design leverages the specific strengths of both approaches: deterministic rules ensure robust coverage of fixed-format identifiers while the neural architecture provides the semantic adaptability required to identify PII within the open-ended, multilingual structures of chatbot conversations. The resulting anonymisation process transforms PII into standardised placeholders, preserving the original conversational hierarchy and ensuring that the data remain suitable for subsequent analytical workflows, such as lexicometric and thematic analyses.

4.5 Error Independence and Statistical Robustness

The hybrid architecture relies on a simplifying assumption of weak dependence between error sources across the pipeline. Errors from the rule-based component and the neural model are assumed to be largely uncorrelated due to the different nature of their mechanism: deterministic pattern matching versus contextual semantic inference.

This assumption should be interpreted as a modelling simplification rather than a strictly verifiable property, as some dependency may arise in ambiguous cases. However, the modular

structure of the pipeline reduces the likelihood of strong error propagation between components.

At the data level, the use of individual conversational exchanges further limits systematic dependencies across observations. Bootstrap-based confidence intervals observed in the evaluation phase are consistent with a stable error distribution, although this is interpreted as descriptive evidence rather than a formal test of independence.

4.6 Statistical Assessment of Residual Disclosure Risk

Despite the high sensitivity of the hybrid pipeline, the removal of explicit identifiers does not eliminate disclosure risk. Indirect identification may still arise from undetected quasi-identifiers or rare patterns (El Emam et al., 2015).

To quantify this, we formalise a conversation-level risk metric. Under the simplifying assumption of independence between token-level errors (as discussed in Section 4.5), the residual disclosure risk R_j for a conversation j can be approximated as the probability of at least one sensitive entity being missed:

$$R_j = 1 - \prod_{i \in j} 1 - \hat{p}_{FN,i} \quad (1)$$

where $\hat{p}_{FN,i}$ denotes the estimated probability of a false negative for the i -th unit. This formulation provides a point estimate of risk but does not account for estimation uncertainty. Therefore, we embed it in a bootstrap-based framework consistent with Section 4.1, obtaining an empirical distribution of R_j through repeated resampling.

This allows us to derive confidence intervals for residual disclosure risk, providing uncertainty-aware estimates of privacy protection. In operational terms, this enables data controllers to assess not only expected risk levels but also their variability under sampling uncertainty. Rather than relying on deterministic thresholds, the framework expresses privacy guarantees in probabilistic terms, in line with inferential practices in official statistics and privacy-by-design principles.

The proposed conversation-level risk indicator can support operational monitoring within the anonymisation workflow. In particular, it can be used as a pre-storage quality control tool to identify conversations or batches of dialogues with elevated residual disclosure risk, thereby enabling targeted inspection or additional anonymisation where necessary.

5. Results and Discussion

The empirical evaluation of the proposed framework was conducted on a manually curated gold standard of 300 conversational records sampled from the two largest daily logs. This dataset reflects the linguistic variability, noise, and heterogeneity typical of operational chatbot environments.

The hybrid pipeline achieves a high Sensitivity (0.9661), indicating that most PII instances are correctly identified despite the presence of non-standard expressions and domain-specific terminology. The bootstrap-based 95% confidence interval [0.9138, 1.0000] confirms the stability of this result across resampling iterations, suggesting robust performance under uncertainty.

Information Loss is very low (0.0017), indicating minimal over-masking of non-sensitive tokens. This suggests that the anonymised text preserves both lexical richness and semantic structure, supporting downstream analytical tasks such as lexicometric and thematic analysis.

Precision (0.7037) indicates a moderate level of over-masking. This outcome reflects a deliberate conservative design choice, where ambiguous entities (e.g. partial addresses or professional references) are more likely to be masked to reduce false negatives. In statistical disclosure control terms, this corresponds to an asymmetric cost structure in which false negatives are penalised more heavily than false positives.

The False Negative Rate (0.0339) provides an upper bound on residual disclosure risk at the token level. Taken together, these metrics highlight an explicit trade-off between privacy protection and data utility, with the system prioritising risk minimisation.

Building on token-level results, we analyse the conversation-level risk indicator introduced in Section 4.6. This measure aggregates false negative probabilities across tokens, capturing the cumulative risk of undetected PII within entire dialogues. The bootstrap-based framework provides confidence intervals for this indicator, enabling uncertainty-aware assessment of disclosure risk at the interaction level.

From an operational perspective, this enables data controllers in institutional contexts to evaluate anonymisation quality in probabilistic terms rather than relying on deterministic thresholds, thereby supporting risk-aware decision-making in line with privacy-by-design principles.

The results show that the proposed hybrid framework achieves a strong balance between detection performance and information preservation. The integration of NLP-based extraction with inferential evaluation enables both effective anonymisation and a transparent quantification of residual risk, consistent with the requirements of official statistical systems.

6. Conclusions

This paper presented a privacy-by-design framework for the anonymisation of multilingual chatbot conversations at ISTAT. The proposed hybrid pipeline demonstrates that effective PII removal can be achieved while preserving the analytical usability of unstructured textual data.

The empirical evaluation on real conversational data confirms the robustness of the approach, showing consistently high detection performance and very low information loss. Overall, the results highlight a stable balance between privacy protection and preservation of textual content, in line with the requirements of official statistical production.

A key contribution of this work is the integration of NLP-based anonymisation techniques within a Statistical Disclosure Control framework. In particular, the adoption of a bootstrap-based inferential evaluation allows performance and residual risk to be interpreted in probabilistic terms, supporting more informed decision-making in operational contexts.

The findings indicate that a conservative calibration of the system is effective in prioritising the minimisation of disclosure risk, while maintaining sufficient data quality for downstream lexicometric and thematic analyses. This design choice is particularly relevant in institutional settings where privacy constraints dominate over marginal gains in data precision.

Despite these encouraging results, some limitations remain. In particular, the use of synthetically generated training data, although necessary for privacy compliance, may not fully

capture the full complexity and variability of real conversational language, which could affect generalisation in more heterogeneous contexts.

Future work will focus on improving the handling of indirect and context-dependent forms of identification, as well as assessing the scalability of the framework across different types of unstructured data within official statistics.

Overall, this study shows that modular hybrid architectures represent a viable and scalable solution for privacy-preserving text analytics in public institutions, enabling the use of advanced NLP techniques while maintaining compliance with stringent data protection requirements.

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